Build Compatibility -- Guided Notes ANSWER KEY

Unit 2.5 | CompTIA Network+ N10-009 Obj. Core 1 3.5 | N10-008

For instructor use / distribute after students complete the notes.

Question 1

The most critical compatibility check in any PC build is the CPU _______. A CPU with socket LGA 1700 _______ (will/will not) fit a motherboard with socket AM4. There is _______ physical adapter or workaround that allows a CPU to be used with an incompatible socket.

Answer: Socket; will NOT; no.

Question 2

DDR4 and DDR5 RAM are physically _______ with each other because the _______ (alignment notch) is in a different position on each generation. Installing a DDR5 stick into a DDR4 slot is _______ without forcing it, which prevents accidental damage. Always verify the motherboard's _______ specification before purchasing RAM.

Answer: Incompatible; key notch; impossible (physically blocked); memory (RAM).

Question 3

GPU physical compatibility requires that the GPU _______ length fits within the case's maximum GPU clearance. A 340mm GPU _______ (will/will not) fit in a case with 310mm GPU clearance. Before buying, check the case _______ for maximum GPU length in millimeters.

Answer: Card; will NOT; specifications (specs / manual).

Question 4

PSU wattage should cover the system's total power requirement plus a _______ % to _______ % headroom buffer. A system with a CPU TDP of 125W, GPU TDP of 200W, drives and fans totaling 50W has an estimated total TDP of _______ W. The recommended minimum PSU for this system would be _______ W.

Answer: 20%; 30%; 375W total TDP; 450-490W PSU (375W x 1.2 to 1.3).

Question 5

A PCIe x16 graphics card _______ (will/will not) fit into a PCIe x1 slot because the x16 card is physically _______ than the x1 slot. However, a PCIe x1 card _______ (will/will not) fit into an x16 slot and will operate at x1 bandwidth. PCIe slots are designed to be electrically _______ compatible.

Answer: Will NOT; longer; will (yes); backward (and forward).

Question 6

An ATX power supply _______ (will/will not) fit in a case designed for SFX form factor. ATX PSUs measure approximately _______ mm long, while SFX PSUs are significantly _______. When building a small form factor (SFF) system, always verify that the PSU form factor matches the _______ specifications.

Answer: Will NOT; 140-160mm; shorter (smaller); case.

Question 7

Before installing a new CPU on an existing motherboard, a technician should consult the manufacturer's ________ to verify the CPU is supported. When a new CPU generation requires a BIOS update for compatibility, the technician must update the BIOS ________ (before/after) installing the new CPU, using a/an ________ CPU that the board already supports.

Answer: CPU support list (QVL); before; compatible (already-supported / older).

Question 8 -- Real World Application

REAL WORLD: A customer brings in a PC build they assembled at home. It will not POST. They have a 13th Gen Intel Core i9 in an LGA 1700 motherboard, 2 x 16 GB DDR5-5600 RAM, and a 4 TB NVMe SSD. The motherboard is an older model manufactured before 13th Gen CPUs were released. List the compatibility checks you would perform in order and explain what you expect to find.

Answer: Step 1 -- Check CPU compatibility list on the manufacturer website for this exact motherboard model. Older LGA 1700 boards support 12th Gen natively; 13th Gen support requires a BIOS update. Step 2 -- Check BIOS version. If the current BIOS is too old to support 13th Gen, the board will not POST. A BIOS update using a 12th Gen CPU would be needed first. Step 3 -- Verify DDR5 slot compatibility -- confirm the board supports DDR5 (not DDR4). Step 4 -- Verify M.2 slot supports NVMe PCIe 4.0 or 5.0 for the SSD. Step 5 -- Check PSU wattage for a Core i9 + high-end SSD (minimum 650W recommended). Most likely cause: BIOS too old to recognize 13th Gen CPU.